

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No- 4	Team No-	Project Title- E-Commerce Product Catalogue
Team Lead Roll Number		2520030524(CSE)
Team Lead Name		J. Advait
Team Member Roll Number		2520090102(CSIT)
Team Member Name		D. Pawan Kumar

1. Abstract:

Online shopping has become an everyday activity for millions of users, yet many e-commerce platforms struggle to provide a smooth and intuitive browsing experience. This project, the E-Commerce Product Catalogue, aims to solve that problem by building a feature-rich, responsive web application that makes product discovery easy and enjoyable.

The application supports multi-level product categories, allowing users to drill down from broad categories to specific subcategories. A powerful filter panel lets shoppers narrow results by price, brand, rating, colour, and size simultaneously. An intelligent search bar offers real-time suggestions as the user types, significantly reducing the time taken to find a desired item. Users can also compare multiple products side by side, helping them make more informed purchase decisions.

The solution is built using React.js on the frontend for a dynamic, component-driven interface, with Node.js and Express.js powering the backend REST APIs. State management is handled through Context API or Redux, and the UI is styled using Bootstrap or Tailwind CSS for a clean, professional look. An optional MySQL database layer supports persistent product and user data storage. The project consists of over ten distinct modules including a product grid, category navigation, filter panel, search bar, sort options, compare tool, product detail view, wishlist, browsing history, and an admin panel for content management. Together, these modules deliver a complete, scalable, and maintainable e-commerce catalogue experience.

2. Module Mapping:

Module Name	Assigned Roll Number(s)	Signature
Product Grid & Category Navigation	2520030524	
Filter Panel, Search Bar & Sort Options	2520030524	
Compare Tool & Product Detail View	2520090102	
Wishlist, History & Admin Panel	2520090102	

3. Module Details:

J. Advait (2520030524) – Responsible for designing and implementing the Product Grid and Category Navigation. This includes building reusable React components for displaying products in a structured grid layout, rendering multi-level category trees, and wiring up all frontend interactions with the backend REST APIs. Also in charge of the Filter Panel, Search Bar, and Sort Options — developing the rich filtering UI (price, brand, rating, colour, size), connecting real-time search suggestions using Axios, and enabling multiple sort orders dynamically.

D. Pawan Kumar (2520090102) – Responsible for building the Compare Tool and Product Detail View, giving users the ability to select and compare multiple products on a single page with key attributes highlighted side by side. Also handles the Wishlist, Browsing History, and Admin Panel modules — implementing LocalStorage-based wishlist persistence, storing and displaying the user's recently viewed items, and creating a fully functional admin interface for managing product listings, categories, and content.

4. Frameworks / Platforms / Software Used:

Frontend: HTML5, CSS3, JavaScript (ES6+), React.js, React Router, Context API / Redux, Bootstrap / Tailwind CSS, Axios.

Backend: Node.js, Express.js, REST APIs.

Database: MySQL (optional).

Testing / Tools: npm, Postman (API Testing).

Guide Name and Signature (remarks if any) _____